Physics 5020: HW12.

(Dated: April 18, 2012)

Problems 1 (1.33) and 2 (1.34). [2 points each]

Problem 1.33. An ideal gas is made to undergo the cyclic process shown in Figure 1.10(a). For each of the steps A, B, and C, determine whether each of the following is positive, negative, or zero: (a) the work done on the gas; (b) the change in the energy content of the gas; (c) the heat added to the gas. Then determine the sign of each of these three quantities for the whole cycle. What does this process accomplish?

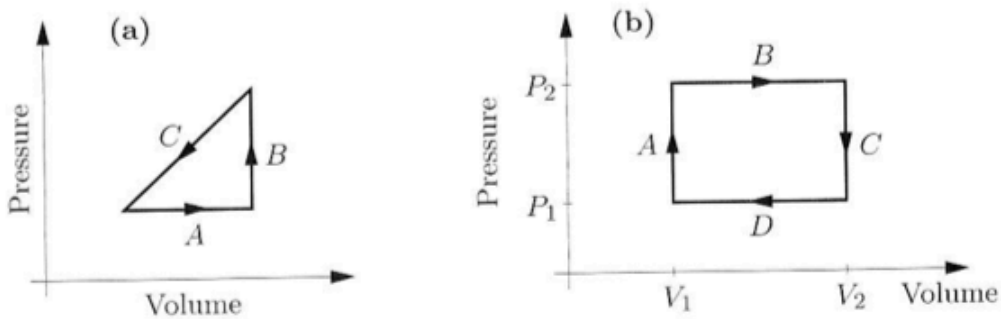

Figure 1.10. PV diagrams for Problems 1.33 and 1.34.

Problem 1.34. An ideal diatomic gas, in a cylinder with a movable piston, undergoes the rectangular cyclic process shown in Figure 1.10(b). Assume that the temperature is always such that rotational degrees of freedom are active, but vibrational modes are "frozen out." Also assume that the only type of work done on the gas is quasistatic compression-expansion work.

(a) For each of the four steps A through D, compute the work done on the gas, the heat added to the gas, and the change in the energy content of the gas. Express all answers in terms of P_1, P_2, V_1, and V_2. (Hint: Compute ΔU before Q, using the ideal gas law and the equipartition theorem.)

(b) Describe in words what is physically being done during each of the four steps; for example, during step A, heat is added to the gas (from an external flame or something) while the piston is held fixed.

(c) Compute the net work done on the gas, the net heat added to the gas, and the net change in the energy of the gas during the entire cycle. Are the results as you expected? Explain briefly.

Problem 3. [3 points]
Calculate the Maxwell distribution and average speed of free particles in *two* dimensions.

Problem 4. [3 points]
Calculate the rate of collisions with unit area of the wall, i.e. the number of molecules striking unit area per unit time, for a classical three-dimensional Maxwell gas at temperature T. Express your result in terms of the particle's concentration $n = N/V$ and the mean speed $\bar{v} = \int_0^\infty v f(v) dv$. *Hint:* see posted lecture note on Maxwell's distribution.